

Competency

- To collect, present and analyze the data

Learning Outcomes

- To construct the frequency distribution table for the collected data
- To construct the histogram, line graph and pie-chart for the collected data and solve the related problems
- To introduce the less and more than ogive and construct them
- To find the mean, median, mode and the quartiles of the ungrouped data.
- To collect the real (primary and secondary) data and analyze the data using the appropriate statistical measure.

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define histogram - To define pie-chart - To recall the formula of finding the mean, median and quartiles on ungrouped data
2.	Understanding (U)	- To answer the questions from histogram and line graph - To construct the histogram to represent the given grouped data - To find the mean, median of individual series
3.	Application (A)	- To represent the data through pie-chart - To calculate the mean, median and quartiles of discrete series - To draw the less and more than ogives and identify the median from the graph
4.	High Ability (HA)	- To collect the real (primary and secondary) data and analyze the data using the appropriate statistical measure and analyze the data

Required Teaching Materials/ Resources

Colourful chart-paper, colourful markers, chart paper with required formulae, graph-board, graph paper, highlighter etc

Pre-knowledge: Frequency distribution table, mean, median etc

Teaching Activities

1. Explain about statistics
2. Discuss on data and types of data with real life examples
3. With example, discuss about the frequency tables
4. Explain with examples about the histogram, its importance and procedures of constructing histogram
5. Divide the students and tell to draw histograms from the questions given in exercise and present in class
6. Ask about the pie-chart, its importance and procedures of its construction.
7. Divide the students and tell to show the data in pie-charts from the questions given in exercise and present in class
8. Discuss about the ogives with examples
9. With appropriate examples, discuss on the central tendencies and their calculations
10. With discussion, list out the following formulae
 - (i) Mean
 - (a) For individual data; $\text{Mean } (\bar{x}) = \frac{\sum x}{n}$
 - (b) For discrete data; $\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f}$
 - (ii) Median
For individual and discrete data; position of median = $\left(\frac{n+1}{2}\right)^{\text{th}}$
 - (iii) Mode
 - (a) For individual and discrete data; mode = item having highest frequency
 - (b) For continuous data; mode = $\text{Mode } (M_0) = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times c$
where L = lower limit of model class, f_0 = frequency of the class preceding to model class, f_1 = frequency of model class, f_2 = frequency of the class succeeding to model class and c = the width of the class
 - (iv) Quartiles
 - (a) For individual and discrete data; position of $Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}}$ term
 - (b) For individual and discrete data; position of $Q_3 = 3 \left(\frac{n+1}{4}\right)^{\text{th}}$ term

Solution of selected problems from Vedanta Excel in Mathematics

- 1) The pie chart given alongside shows the votes secured by three candidates X, Y and Z in an election. If X secured 5760 votes,

- i) how many votes did Z secure?
 ii) who secured the least number of votes? How many votes did he secure?

Solution:

Let the total number of votes secured by three candidates be x .

Then, the number of votes secured by $x = 5760$

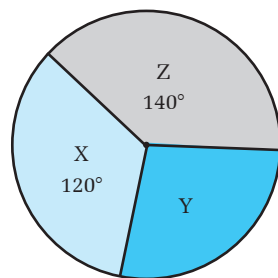
$$\text{or, } \frac{120}{360} \times x = 5760$$

$$\therefore x = 17280$$

i) No. of votes secured by Z = $\frac{140}{360} \times 17280 = 6720$

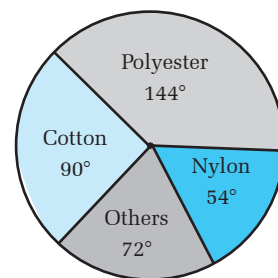
ii) No. of votes secured by Y = $\frac{360 - (120 + 140)}{360} \times 17280$
 $= \frac{140}{360} \times 17280 = 4800$

Y secured the least number of votes. He secured only 4800 votes.



- 2) The given pie chart shows the composition of different materials in a type of cloth in percentage.

- i) Calculate the percentage of each material found in the cloth.
 ii) Calculate the weight of each material contained by a bundle of 50 kg of cloth.



Solution:

Materials	In Percentage	Weight (in kg)
Polyster	$\frac{144}{360} \times 100 \% = 40 \%$	40 % of 50 kg = 20 kg
Cotton	$\frac{90}{360} \times 100 \% = 25 \%$	25 % of 50 kg = 12.5 kg
Nylon	$\frac{54}{360} \times 100 \% = 15 \%$	15 % of 50 kg = 7.5 kg
Other	$\frac{72}{360} \times 100 \% = 20 \%$	20 % of 50 kg = 10 kg

3. Draw a 'less than' ogive from the data given below.

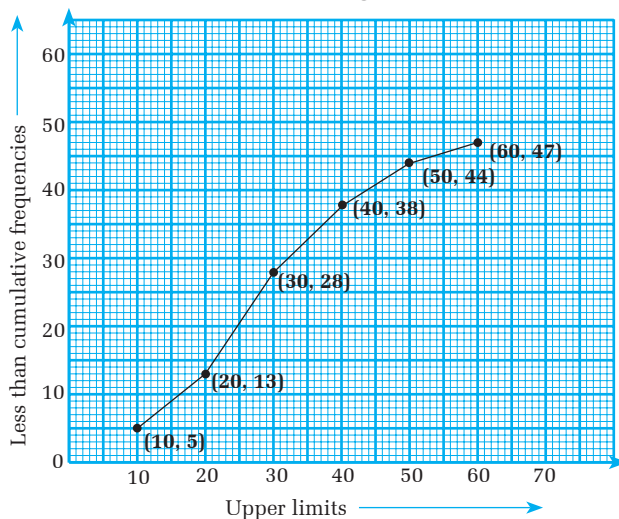
Data	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency	5	8	15	10	6	3

Solution:

Less than cumulative frequency table.

Marks	No. of students (f)	Upper limit	less than c.f.	Coordinates (x, y)
0 – 10	5	10	5	(10, 5)
10 – 20	8	20	13	(20, 13)
20 – 30	15	30	28	(30, 28)
30 – 40	10	40	38	(40, 38)
40 – 50	6	50	44	(50, 44)
50 – 60	3	60	47	(60, 47)

less than ogive



4. The speeds of vehicles recorded in a highway during 15 minutes on a day is given in the following table. The average speed of the vehicles was 54 km per hour.

Speed (km/hr)	10	30	50	70	90
No. of vehicles	7	k	10	9	13

- (i) Find the value of k. (ii) Find the total number of vehicles counted during the time.

Solution:

Speed (km/hr)	No. of vehicles (f)	fx
10	7	70
30	k	30k
50	10	500
70	9	630
90	13	1170
Total	N = 39 + k	$\Sigma fx = 2370 + 30k$

$$\text{Now, mean } (\bar{x}) = \frac{\Sigma fx}{N}$$

$$\text{or, } 54 = \frac{2370 + 30k}{39 + k}$$

$$\text{or, } 2106 + 54k = 2370 + 30k$$

$$\text{or, } 24k = 264$$

$$\therefore k = 11$$

- i) Required value of k is 11.
 ii) No. of vehicles counted during the time is $39 + k = 39 + 11 = 50$

5. Given that mean is 40 and $N = 51$, find the missing frequencies in the following data.

x	10	20	30	40	50	60
f	2	3	-	21	-	5

Solution:

Let the missing frequencies be a and b .

x	No. of vehicles (f)	fx
10	2	20
20	3	60
30	a	$30a$
40	21	840
50	b	$50b$
60	5	300
Total	$N = 31 + a + b$	$\Sigma fx = 1220 + 30a + 50b$

Now,

$$N = 51$$

$$\text{or, } 31 + a + b = 51 \quad \therefore a = 20 - b \dots\dots\dots (i)$$

Again,

$$\text{mean } (\bar{x}) = \frac{\Sigma fx}{N}$$

$$\text{or, } 40 = \frac{1220 + 30a + 50b}{51}$$

$$\text{or, } 2040 = 1220 + 30a + 50b$$

$$\text{or, } 820 = 30a + 50b$$

$$\text{or, } 82 = 3a + 5b \dots\dots\dots (ii)$$

Substituting the value of a from equation (i) in equation (ii), we get

$$82 = 3(20 - b) + 5b$$

$$\text{or, } 22 = 2b \quad \therefore b = 11$$

Putting the value of b in equation (i), we get

$$a = 20 - 11 = 9$$

Hence, the missing frequencies are 9 and 11.

6. Find the first quartile (Q_1) and third quartile (Q_3) from the following distribution.

Age (in year)	22	27	32	37	42
No. of people	35	42	40	30	24

Solution:

Cumulative frequency distribution table,

Age (in years)	No. of people (f)	c.f.
22	35	35
27	42	77
32	40	117
37	30	147
42	24	171
Total	$N = 171$	

Now,

$$\text{Position of } (Q_1) = \left(\frac{N+1}{4}\right)^{\text{th}} \text{ term} = \left(\frac{171+1}{4}\right)^{\text{th}} \text{ term} = 43^{\text{th}} \text{ term}$$

In c.f. column, the c.f. just greater than 43 is 77 and corresponding value is 27.

$\therefore$ The first quartile (Q_1) = 27

Also,

$$\text{Position of median } (Q_2) = \left(\frac{N+1}{2}\right)^{\text{th}} \text{ term} = \left(\frac{171+1}{2}\right)^{\text{th}} \text{ term} = 86^{\text{th}} \text{ term}$$

In c.f. column, the c.f. just greater than 86 is 117 and its corresponding value is 32.

$\therefore$ Median (Q_2) = 32

Again,

$$\text{Position of } (Q_3) = \frac{3(N+1)}{4}^{\text{th}} \text{ term} = 129^{\text{th}} \text{ term}$$

In c.f. column, the c.f. just greater than 129 is 147 and its corresponding value is 37.

$\therefore$ The third quartile (Q_3) = 37

Extra Question

1. Construct a histogram from the data given in the table below.

Marks	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
No. of student	4	7	8	5	3

2. A mobile shop sold the mobile sets of the following brands in a month. Draw a pie chart to show the data.

Brand	No. of mobile sets
Samsung	130
Huawei	120
Oppo	90
Vivo	40
Nokia	20

3. Draw a 'less than' ogive from the data given below.

Wages (in Rs)	400 - 500	500 - 600	600 - 700	700 - 800	800 - 900
No. of workers	4	3	6	7	5

4. The mean of the data given below is 17. Determine the value of m.

X	5	10	15	20	25	30
f	2	5	10	m	4	2

[Ans: 7]

5. Find the quartiles from the following distribution.

Height (in cm)	90	100	110	120	130	140
No. of students	20	28	24	40	35	18

[Ans: $Q_1 = 100$, $Q_2 = 120$, $Q_3 = 130$]